



《多模态大模型》

Lecture 15

大模型和Agent Skill: 实战与经验

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本节内容

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一、**AI模型实战**

二、Agent Skill实战

AI模型实战经验

□此部分内容约**50页slides**，此处略

本节内容

CONTENTS

- 一、AI模型实战
- 二、Agent Skill实战

Codex使用skill

我会把刚才的评分表做成一个可填写的 Excel 文件，包含基本信息区、评分项目、得分列、总分和备注区。

已思考 39s >

已做好 Excel 评分表，包含基本信息、评分项目、得分栏、总分公式、等级参考和备注区。

	A	B
1	课堂展示评分	
2		
3	课程名称:	
4	小组编号:	
5	展示主题:	
6		
7	评分项目	分值
8	选题价值与启发性	15 选题与生成式人工智能密切相关; 要问题或发展趋势。
9	内容准确性与完整性	20 基本概念准确; 技术路线、应用背

+ 有问题，尽管问

高级 v



ChatGPT 也可能会犯错。请核查重要信息。

思考

</> 读取SKILL.md文件

</> Python



```
bash -lc cat /home/oai/skills/spreadsheets/  
  
IMPORTANT: INSTRUCTIONS IN THE  
system and user messages ALWAYS  
take precedence over this skill  
---
```

</> 读取API快速入门指南

</> Python



```
bash -lc cat /home/oai/skills/spreadsheets/  
  
'col_offset_px', 'width_px',  
'height_px').  
  
## Discovery Policy (Strict)
```

• 创建工作簿与格式设置

我需要用 artifact_tool 来创建工作簿，而不是 openpyxl。可能会涉及 Python 和分析工具，可能需要检查 API 是否安装。我计划做一个评分表，包含公式和数据验证，还可能应用条件格式化。最终用户可能需要一个 .xlsx 文件。

GPT-5.5

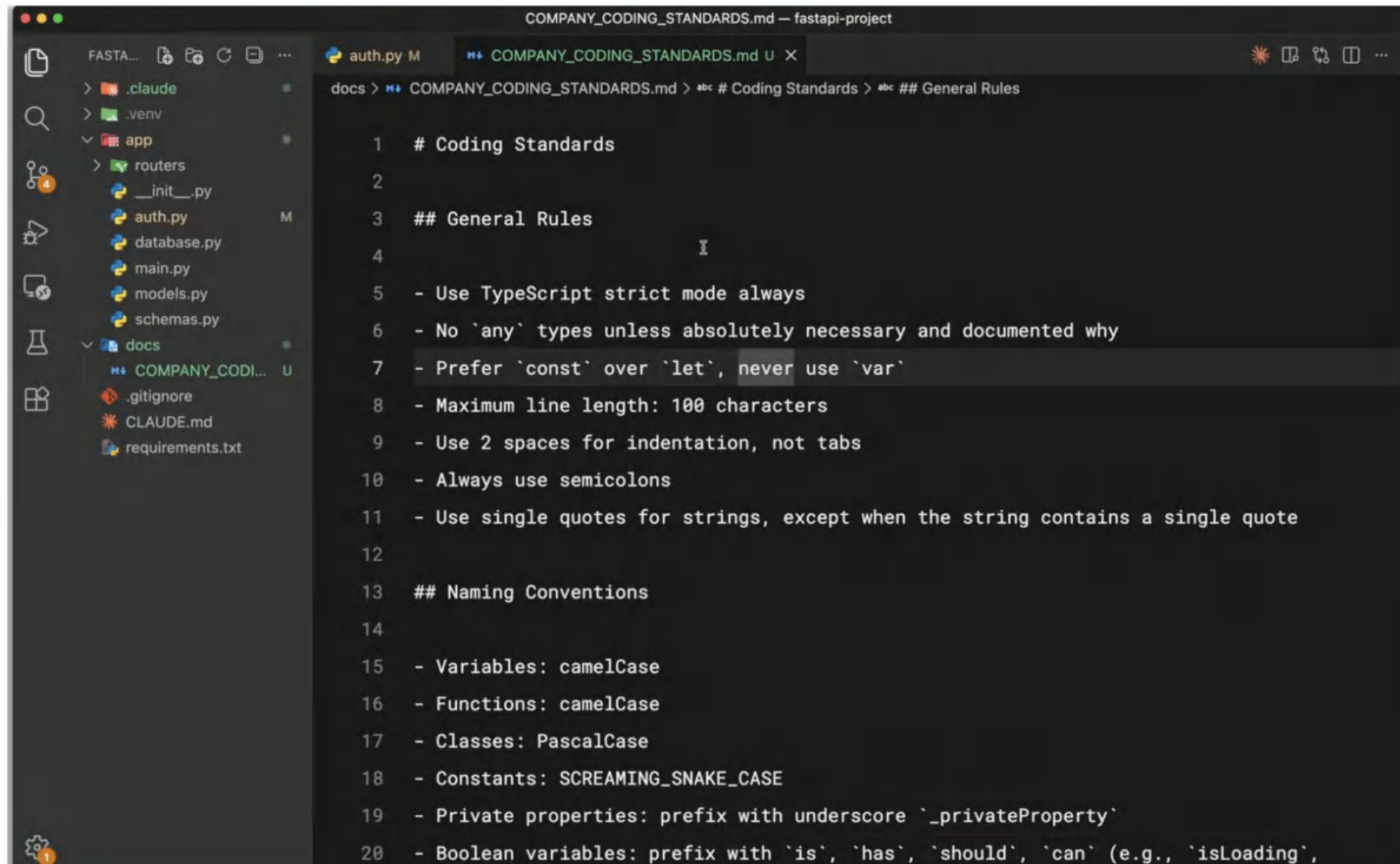
□因此，严格地说，不应该说：

□> **GPT-5.5** 在某任务上成功率是多少。

□而应该说：

□> **GPT-5.5 + Skill X + Codex harness + 某工具环境
+ 某 budget** 的成功率是多少。

如果不使用skill，你要这么做



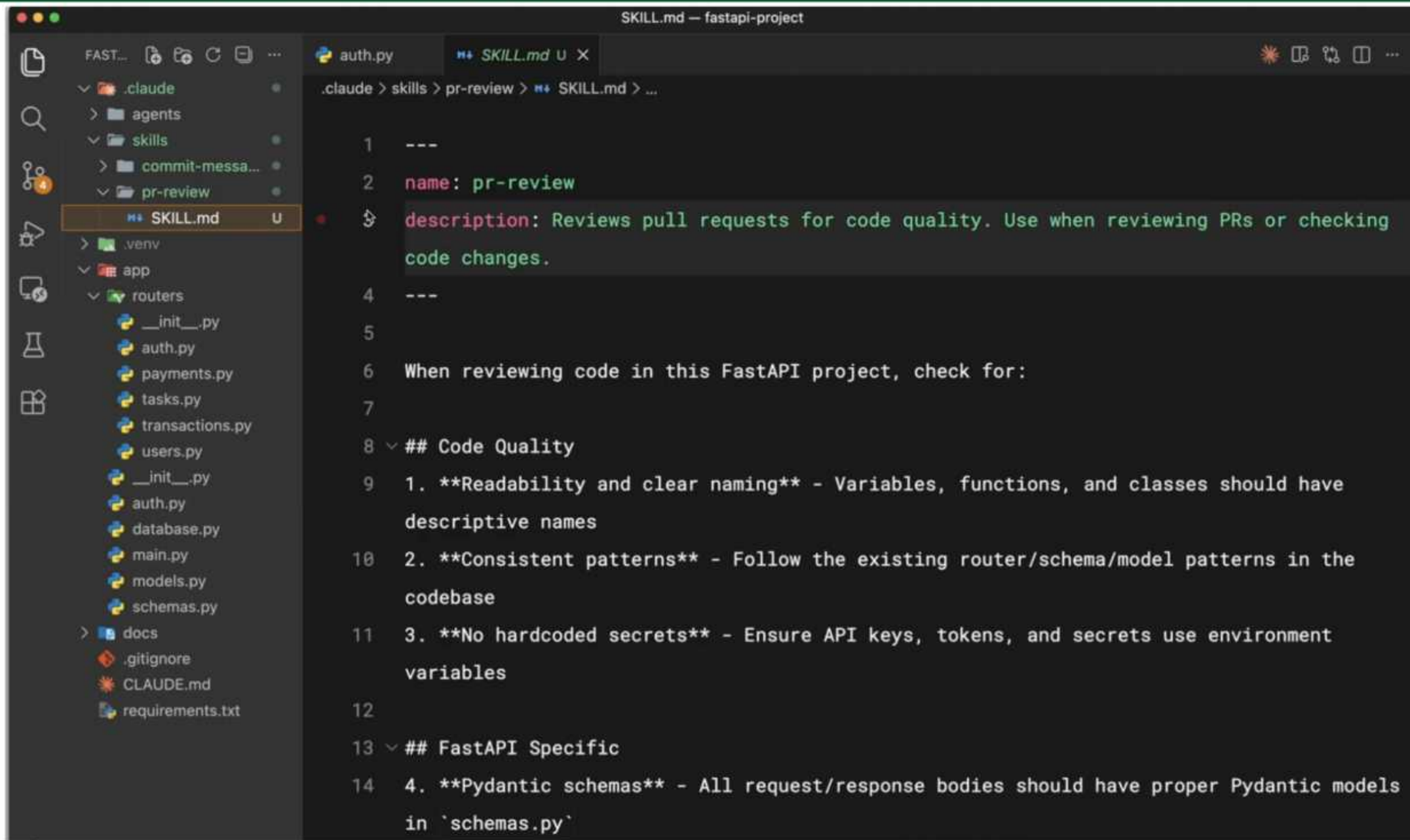
The image shows a screenshot of a VS Code editor window. The title bar reads "COMPANY_CODING_STANDARDS.md — fastapi-project". The left sidebar displays a file explorer with a project structure: "FASTA..." (expanded), ".claude", ".venv", "app" (expanded), "routers" (expanded), and "docs" (expanded). The "docs" folder contains "COMPANY_CODI..." (selected). The main editor area shows the content of "COMPANY_CODING_STANDARDS.md". The breadcrumb navigation at the top of the editor reads "docs > COMPANY_CODING_STANDARDS.md > # Coding Standards > ## General Rules". The code content is as follows:

```
1 # Coding Standards
2
3 ## General Rules
4
5 - Use TypeScript strict mode always
6 - No `any` types unless absolutely necessary and documented why
7 - Prefer `const` over `let`, never use `var`
8 - Maximum line length: 100 characters
9 - Use 2 spaces for indentation, not tabs
10 - Always use semicolons
11 - Use single quotes for strings, except when the string contains a single quote
12
13 ## Naming Conventions
14
15 - Variables: camelCase
16 - Functions: camelCase
17 - Classes: PascalCase
18 - Constants: SCREAMING_SNAKE_CASE
19 - Private properties: prefix with underscore `_privateProperty`
20 - Boolean variables: prefix with `is`, `has`, `should`, `can` (e.g., `isLoading`,
```

一个字一个字地敲

branch. Remember, no emojis or verbose

Skill的位置和内容



```
SKILL.md — fastapi-project

.claude > skills > pr-review > SKILL.md > ...

1 ---
2 name: pr-review
3 description: Reviews pull requests for code quality. Use when reviewing PRs or checking
  code changes.
4 ---
5
6 When reviewing code in this FastAPI project, check for:
7
8 ## Code Quality
9 1. Readability and clear naming - Variables, functions, and classes should have
  descriptive names
10 2. Consistent patterns - Follow the existing router/schema/model patterns in the
  codebase
11 3. No hardcoded secrets - Ensure API keys, tokens, and secrets use environment
  variables
12
13 ## FastAPI Specific
14 4. Pydantic schemas - All request/response bodies should have proper Pydantic models
  in `schemas.py`
```

大模型加载skill

```
> Can you review the pull request on branch sg-221?
```

```
● Skill(pr-review)
```

```
  | Successfully loaded skill
```

```
+ Beboppin'... (esc to interrupt · thinking)
```

```
> █
```

```
  ? for shortcuts
```

加载成功

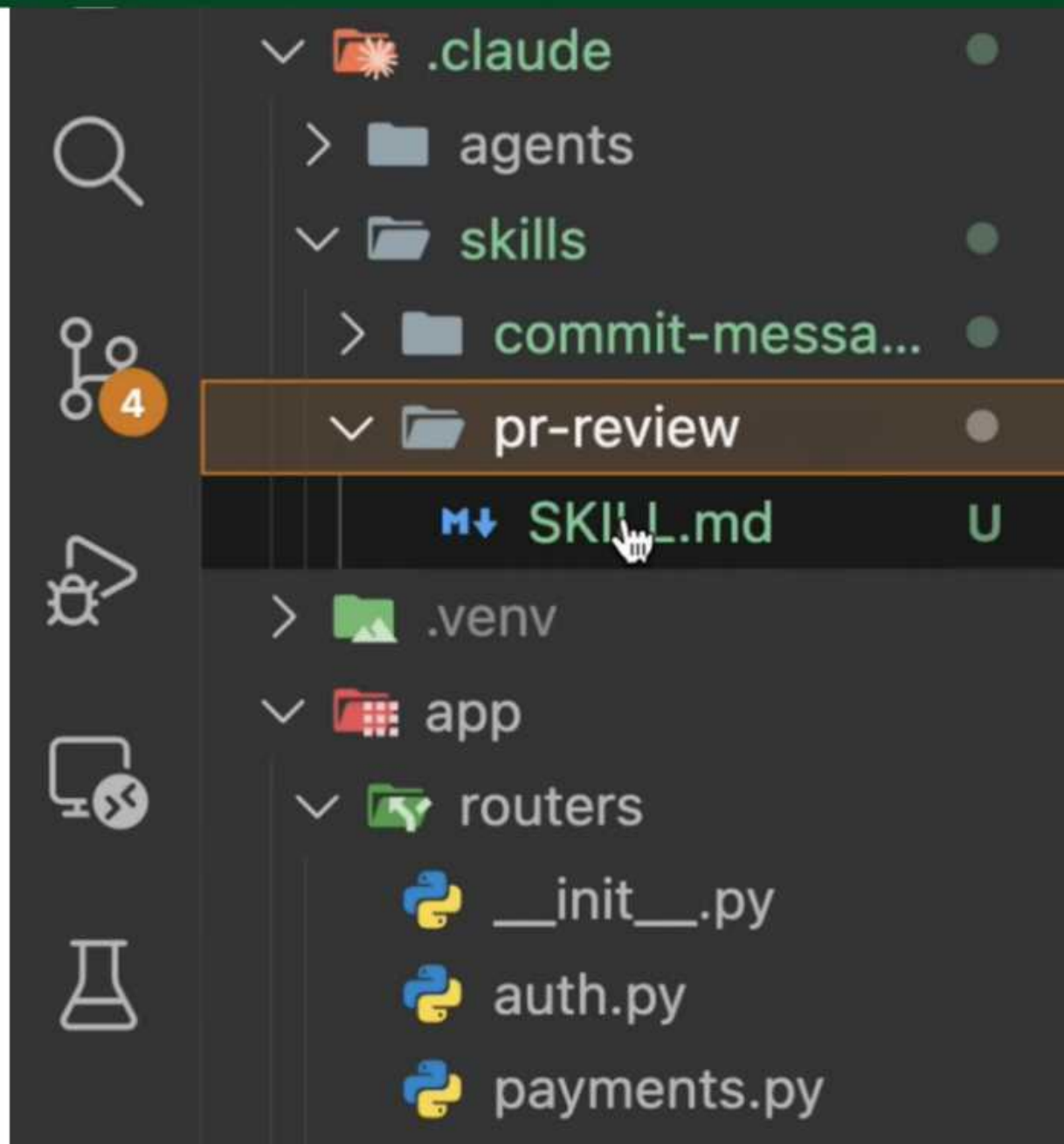
Can you review the pull request

- **Skill**(pr-review)

- L Successfully loaded skill

- * Beboppin'... (esc to interrupt · t

Skill的目录



app > auth.py > ...

15 SECRET_KEY

16 ALGORITHM

17 ACCESS_TOKEN

18

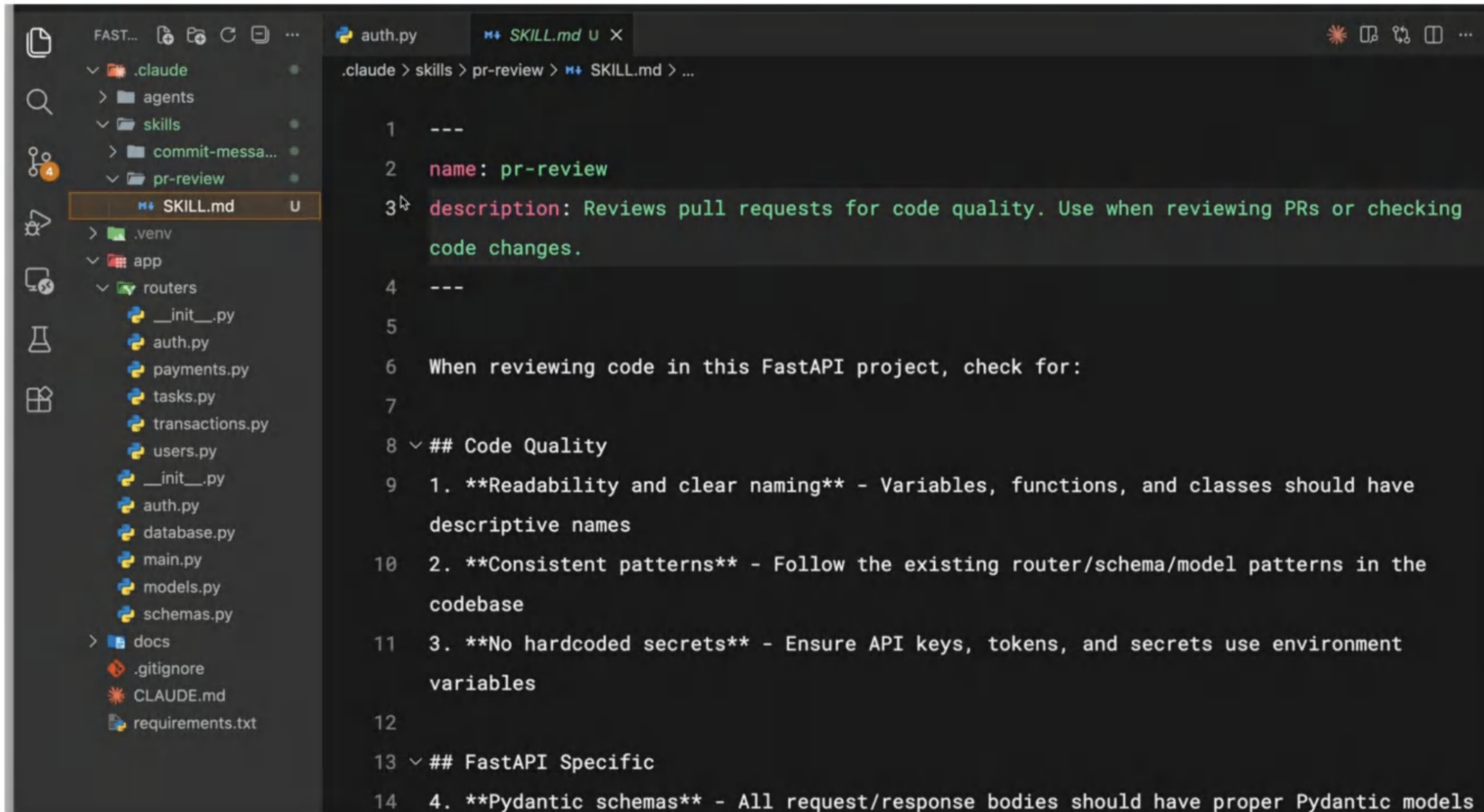
19 pwd_context

20 oauth2_sch

21

22

Skill的内容



```
FAST... .claude agents skills commit-messa... pr-review SKILL.md U .venv app routers __init__.py auth.py payments.py tasks.py transactions.py users.py __init__.py auth.py database.py main.py models.py schemas.py docs .gitignore CLAUDE.md requirements.txt
```

```
auth.py SKILL.md X
```

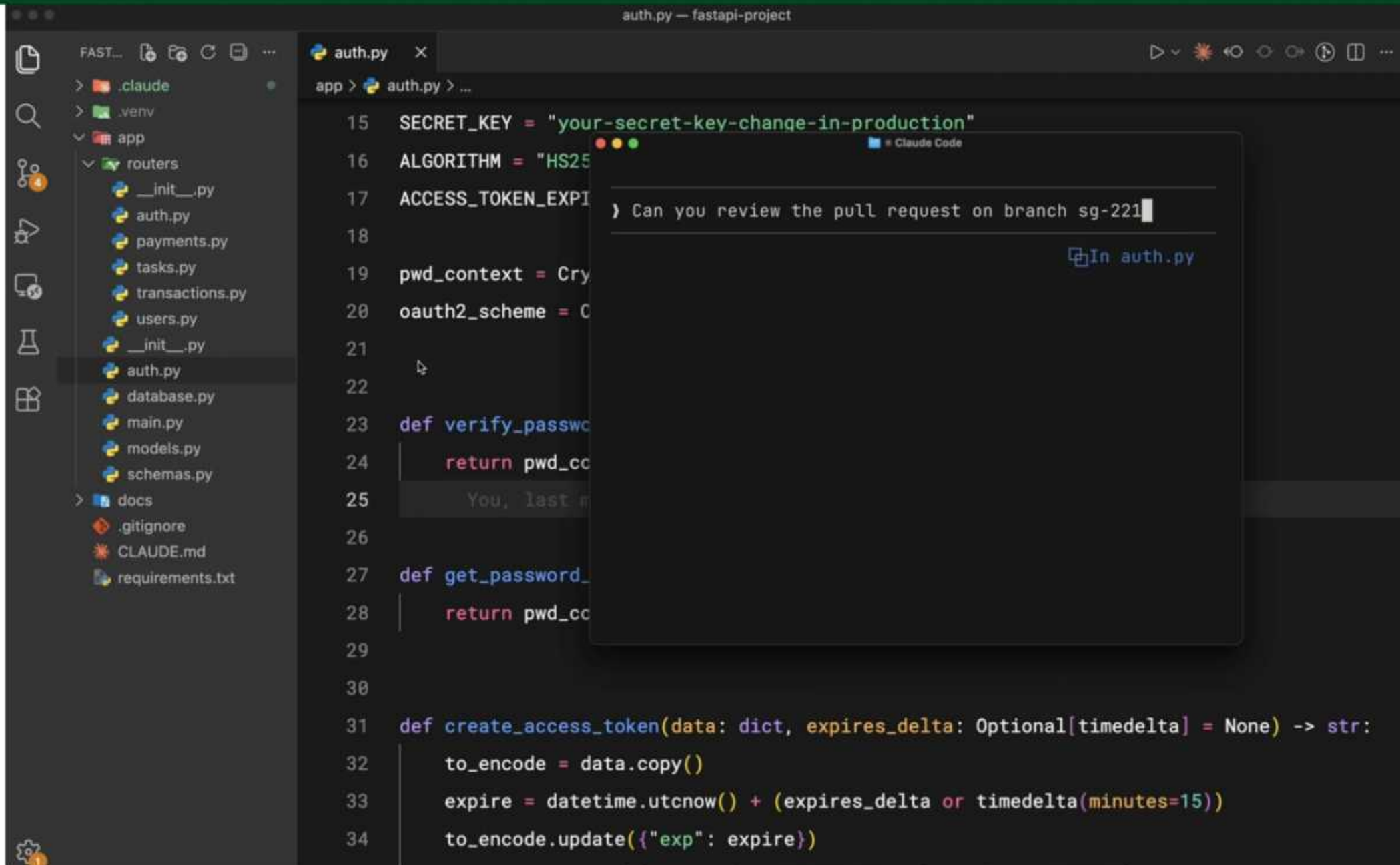
```
.claude > skills > pr-review > SKILL.md > ...
```

```
1 ---
2 name: pr-review
3 description: Reviews pull requests for code quality. Use when reviewing PRs or checking
  code changes.
4 ---
5
6 When reviewing code in this FastAPI project, check for:
7
8 ## Code Quality
9 1. Readability and clear naming - Variables, functions, and classes should have
  descriptive names
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  codebase
11 3. No hardcoded secrets - Ensure API keys, tokens, and secrets use environment
  variables
12
13 ## FastAPI Specific
14 4. Pydantic schemas - All request/response bodies should have proper Pydantic models
```

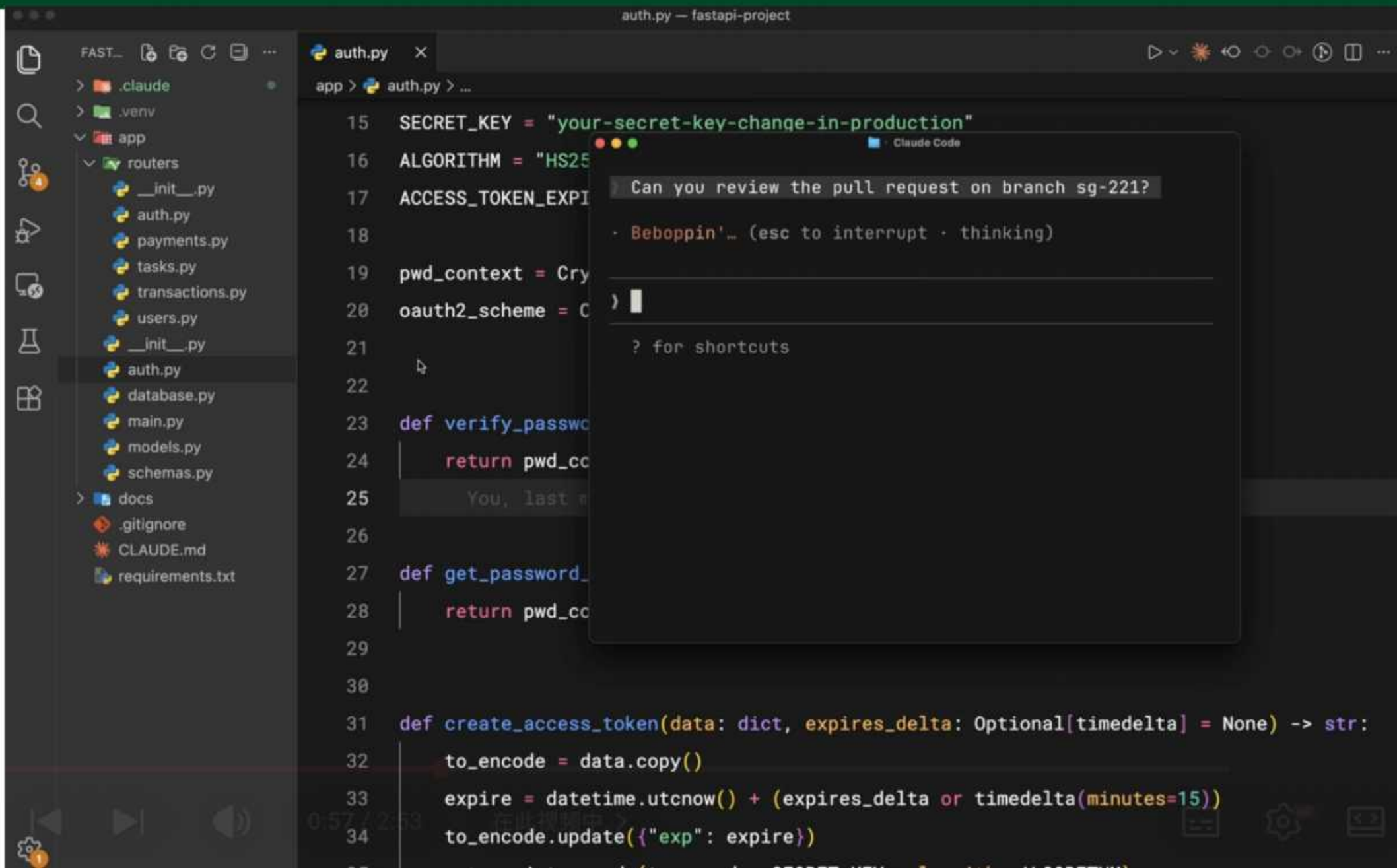
Skill要有一个好的描述

```
2 name: pr-review
3 description: Reviews pull requests for code quality. Use when reviewing PRs or checking
  code changes.
4 ---
```

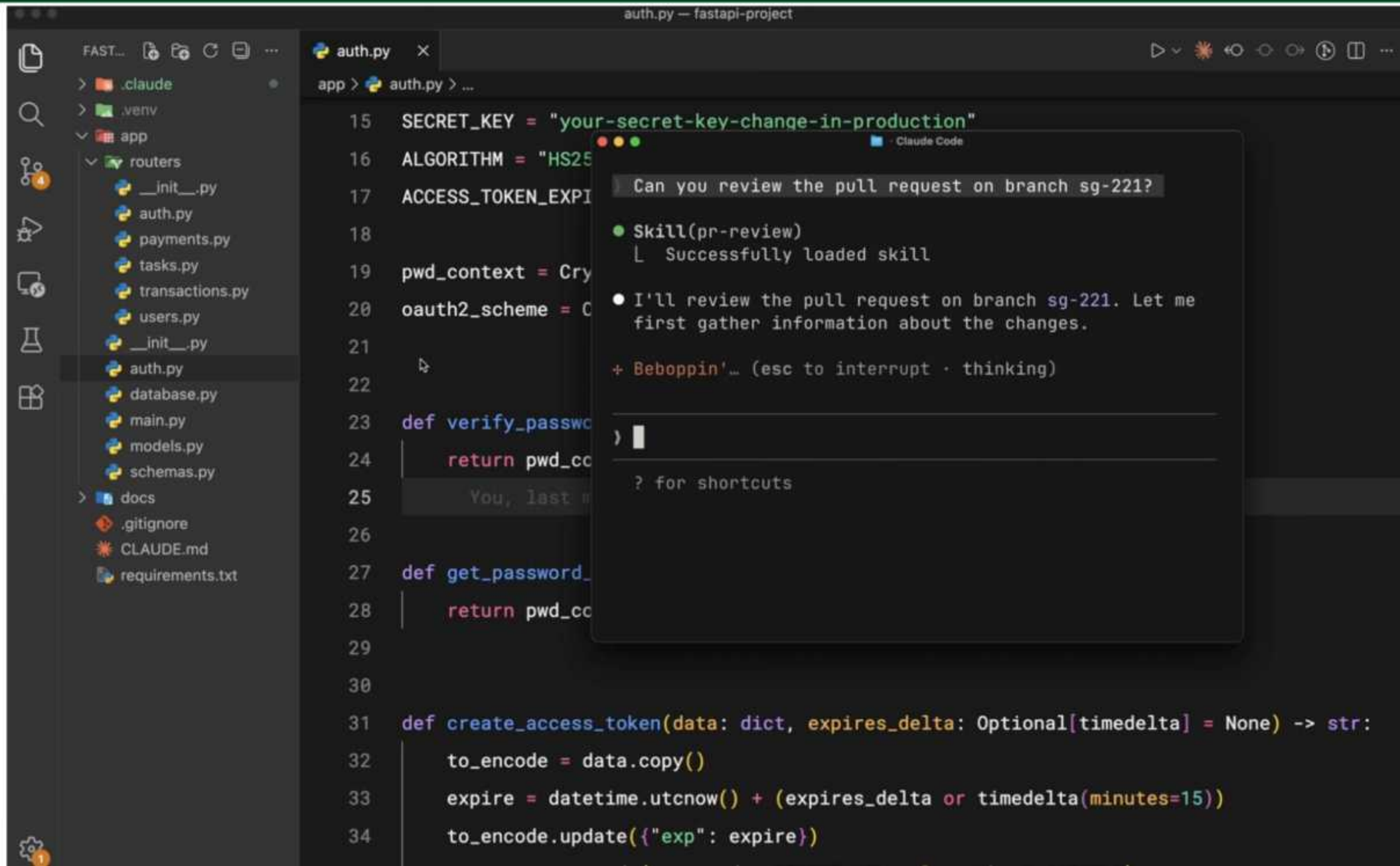
启动Skill



响应中



成功加载



执行Skill

The screenshot displays the Claude Code interface with a file explorer on the left and a code editor in the center. The file explorer shows a project structure with folders like `.claude`, `.venv`, and `app`. The `app` folder contains subfolders like `routers` and `docs`, with files such as `__init__.py`, `auth.py`, `payments.py`, `tasks.py`, `transactions.py`, `users.py`, `database.py`, `main.py`, `models.py`, `schemas.py`, `.gitignore`, `CLAUDE.md`, and `requirements.txt`. The code editor shows the `auth.py` file with the following code:

```
15 SECRET_KEY = "your-secret-key-change-in-production"
16 ALGORITHM = "HS256"
17 ACCESS_TOKEN_EXPIRE_SECONDS = 300
18
19 pwd_context = Cryptor(password_context="pbkdf2:sha256:128")
20 oauth2_scheme = OAuth2PasswordBearer(tokenUrl="/api/auth/login")
21
22
23 def verify_password(password: str, hashed_password: str) -> bool:
24     return pwd_context.verify(password, hashed_password)
25
26 You, last m
27
28 def get_password_hash(password: str) -> str:
29     return pwd_context.hash(password)
30
31
32 def create_access_token(data: dict, expires_delta: Optional[timedelta] = None) -> str:
33     to_encode = data.copy()
34     expire = datetime.utcnow() + (expires_delta or timedelta(minutes=15))
35     to_encode.update({"exp": expire})
```

Overlaid on the code editor is a list of skills:

- Skill(pr-review)
 - └ Successfully loaded skill
- I'll review the pull request on branch `sg-221`. Let me first gather information about the changes.
- Bash(`git fetch origin sg-221 2>/dev/null || git fetch --all 2>/dev/null; git log origin/master..origin/sg-221 --oneline 2>/dev/null || git log master..sg-221 --oneli...`)
 - └ Waiting...
- ★ Beboppin'... (esc to interrupt · thinking)

At the bottom of the skill list, there is a prompt: `? for shortcuts`.

Skill的位置

```
> cd .claude/skills
```

```
> ls
```

```
└─ .
```

```
├─ commit-message
```

```
├─ debugging
```

```
├─ documentation
```

```
└─ pr-description
```

```
> cd commit-message
```

```
> ls
```

```
└─ .
```

```
└─ SKILL.md
```

```
🍏 ~/.claude/skills/commit-message
```

```
>
```

🕒 11:51:51 AM

Github

The screenshot shows a web browser window displaying a GitHub repository page. The address bar shows the URL: `https://github.com/elebumm/fastapi-project/tree/master/.claude/skills`. The repository name is `elebumm / fastapi-project`. The navigation bar includes links for `Code`, `Issues`, `Pull requests` (with a badge of 1), `Actions`, `Projects`, `Security`, `Insights`, and `Settings`. The breadcrumb path is `master` / `fastapi-project` / `.claude` / `skills`. A commit by `elebumm` is shown with the message `docs: add README and PR review guidelines` and hash `dc4aa54`. Below this, a table lists the files in the `skills` directory.

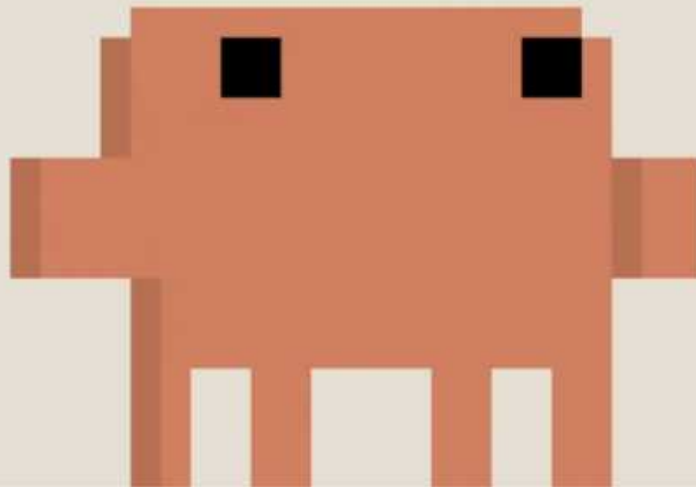
Name	Last commit message	Last commit date
..		
commit-message	docs: add README and PR review guidelines	now
pr-review	docs: add README and PR review guidelines	now

Skill.md vs Claude.md

SKILL.md



CLAUDE.md



加载

HS25

EXPI

Cry

= 0

SSWC

d_co

st m

› Can you review the pull request on branch sg-221?

- Skill(pr-review)

 - └ Successfully loaded skill

- I'll review the pull request on branch sg-221. Let me first gather information about the changes.

+ Beboppin'... (esc to interrupt · thinking)

› █

? for shortcuts

Skill的内容

```
1  ---
2  name: commit-message
3  description: Formats commit messages following project conventions. Useful
   commits or writing commit messages.
4  ---
5
6  Write commit messages following this format:
7
8  ## Structure
9  ```
10 <type>: <short description>
11
12 <optional body explaining why, not what>
```

Skill的描述

```
.claude > skills > pr-description > SKILL.md > ...

You, 14 seconds ago | 1 author (You)

1 ---
2 name: pr-description
3 description: Writes pull request descriptions. Use
4 ---
5
6 When writing a PR description:
7
8 1. Run `git diff main...HEAD` to see all changes on this branch
9 2. Write a description following this format:
10
11 ## What
12 One sentence explaining what this PR does.
13
14 ## Why
15 Brief context on why this change is needed.
16
17 ## Changes
18 - Bullet points of specific changes made
19 - Group related changes together
20 - Mention any files deleted or renamed
21
22 ## Testing
23 How to verify this works. Include specific commands if relevant.
24
25 Keep descriptions concise. Focus on what a reviewer needs to know.
```

Skill的描述

```
# Too similar - Claude can't distinguish
```

```
# Skill 1
```

```
description: Helps with data analysis
```

```
# Skill 2
```

```
description: Analyzes data and generates reports
```

Skill.md vs Claude.md

CLAUDE.md

Project wide standards that always apply

Constraints like "never modify the database schema"

Framework preferences and coding style

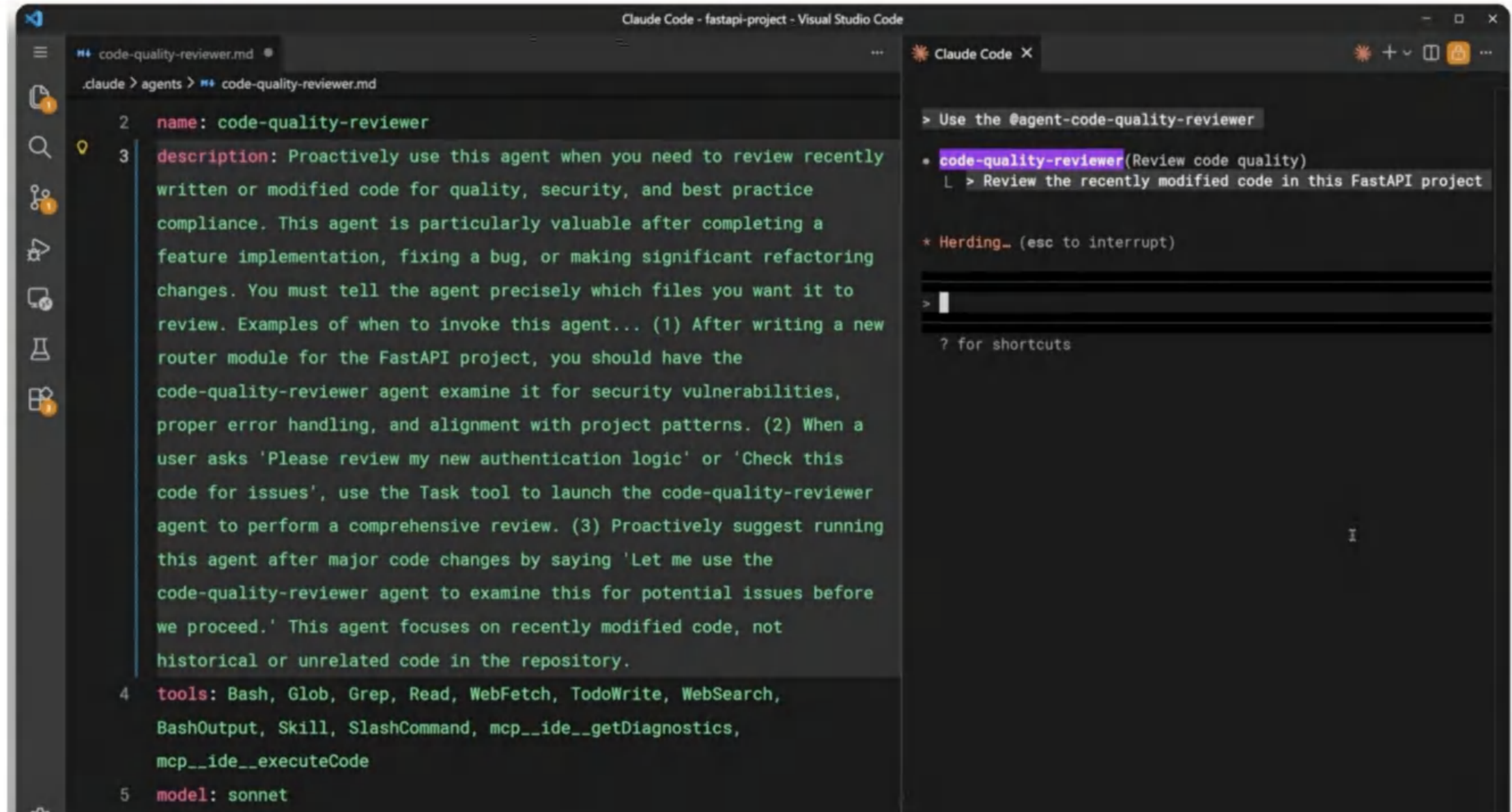
Skills

Task-specific expertise

Knowledge that's only relevant sometimes

Detailed procedures that would clutter every conversation

sub agent



sub agent

ely use this agent when you need to review recently
ode for quality, security, and best practice
t is particularly valuable after completing a
n, fixing a bug, or making significant refactoring
l the agent precisely which files you want it to
hen to invoke this agent... (1) After writing a new
FastAPI project, you should have the
agent examine it for security vulnerabilities,
, and alignment with project patterns. (2) When a
view my new authentication logic' or 'Check this
the Task tool to launch the code-quality-reviewer
prehensive review. (3) Proactively suggest running
r code changes by saying 'Let me use the
agent to examine this for potential issues before
nt focuses on recently modified code, not
ed code in the repository.
ep, Read, WebFetch, TodoWrite, WebSearch,
ashCommand, mcp__ide__getDiagnostics,

> /clear

L (no content)

> Use the @agent-code-quality-reviewer

code-quality-reviewer(Review code quality)

L **Prompt:**

Review the recently modified code in this FastAPI project
for code quality, security, and best practices.

Based on the git status, the modified file is:

- app/routers/users.py

Please perform a comprehensive code review focusing on:

1. Security vulnerabilities (especially important for user-related endpoints)
2. Code quality and adherence to FastAPI/Pydantic best practices
3. Error handling
4. Input validation
5. Any potential issues with the authentication/authorization logic
6. Alignment with the project patterns described in CLAUDE.md

Read the modified file and provide a detailed review with specific recommendations.

> Review the recently modified code in this FastAPI project
ctrl+b to run in background

Subagent vs Skills

Subagents

You want to delegate a task to a separate execution context

You need different tool access than the main conversation

You want isolation between delegated work and main context

Skills

You want to enhance Claude's knowledge for the current task

The expertise applies throughout a conversation

Hooks vs Skills

Hooks

Operations that should run on every file save

Validation before specific tool calls

Automated side effects of Claude's actions

Skills

Knowledge that informs how Claude handles requests

Skill的内容

With optional fields:

```
---  
name: pdf-processing  
description: Extract text and tables from PDF files, fill forms, merge documents.  
license: Apache-2.0  
metadata:  
  author: example-org  
  version: "1.0"  
---
```

Field	Required	Constraints
name	Yes	Max 64 characters. Lowercase letters, numbers, and hyphens only. Must not start or end with a hyphen.
description	Yes	Max 1024 characters. Non-empty. Describes what the skill does and when to use it.
license	No	License name or reference to a bundled license file.
compatibility	No	Max 500 characters. Indicates environment requirements (intended product, system packages, network access, etc.).

Skill的内容

```
description: Extract text and tables from PDF files, fill forms, merge documents.  
license: Apache-2.0  
metadata:  
  author: example-org  
  version: "1.0"  
---
```

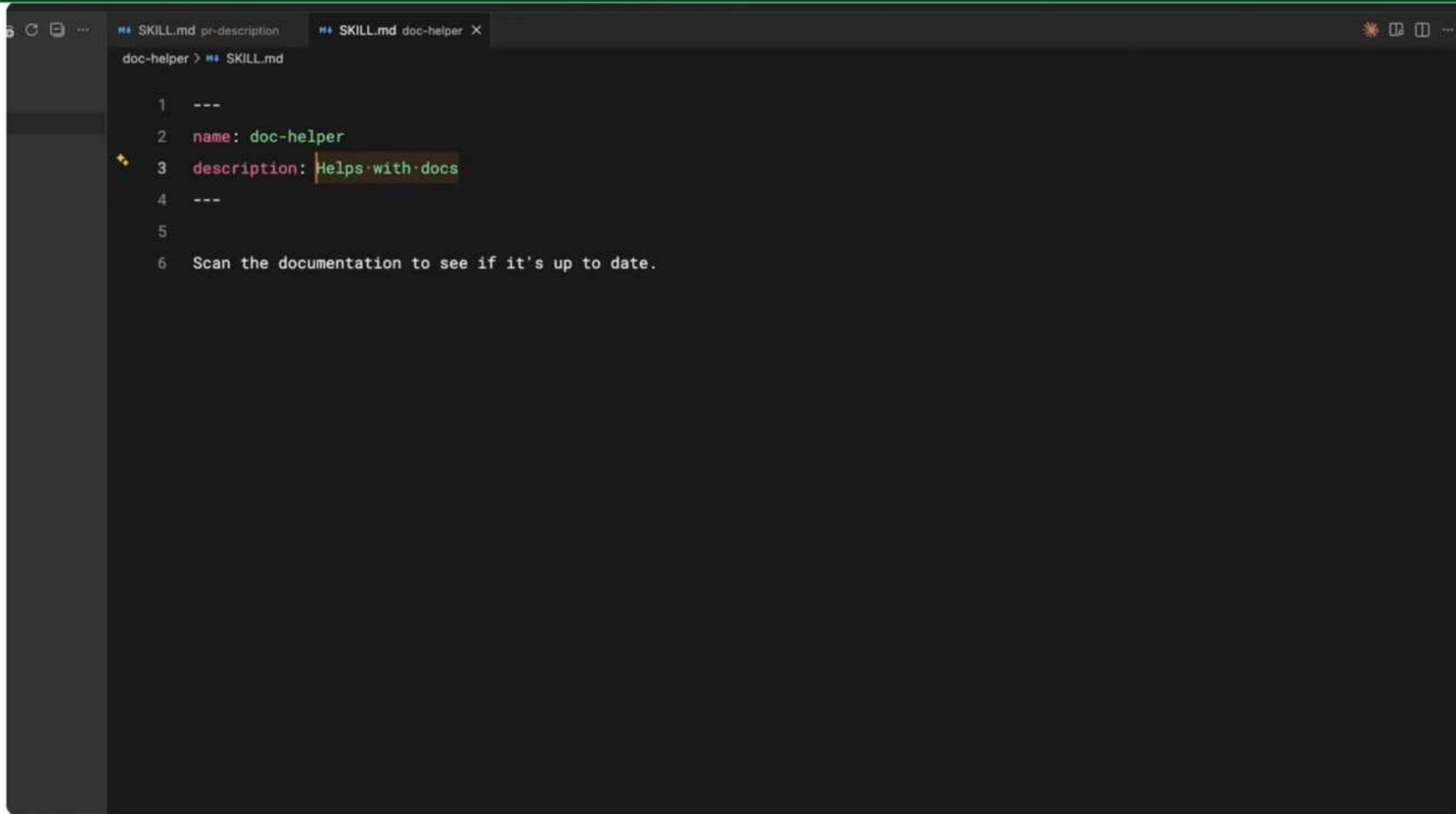
Field	Required	Constraints
name	Yes	Max 64 characters. Lowercase letters, numbers, and hyphens only. Must not start or end with a hyphen.
description	Yes	Max 1024 characters. Non-empty. Describes what the skill does and when to use it.
license	No	License name or reference to a bundled license file.
compatibility	No	Max 500 characters. Indicates environment requirements (intended product, system packages, network access, etc.).
metadata	No	Arbitrary key-value mapping for additional metadata.
allowed-tools	No	Space-delimited list of pre-approved tools the skill may use. (Experimental)

Skill的内容

.claude > skills > code-onboarding > SKILL.md > abc # Codebase Guide > abc ## Quick Start

```
1  ---
2  name: codebase-onboarding
3  description: Helps new developers understand thi
4  the system works.
5  allowed-tools: Read, Grep, Glob, Bash
6  model: sonnet
7  ---
8  ∨ # Codebase Guide
```

Skill的描述

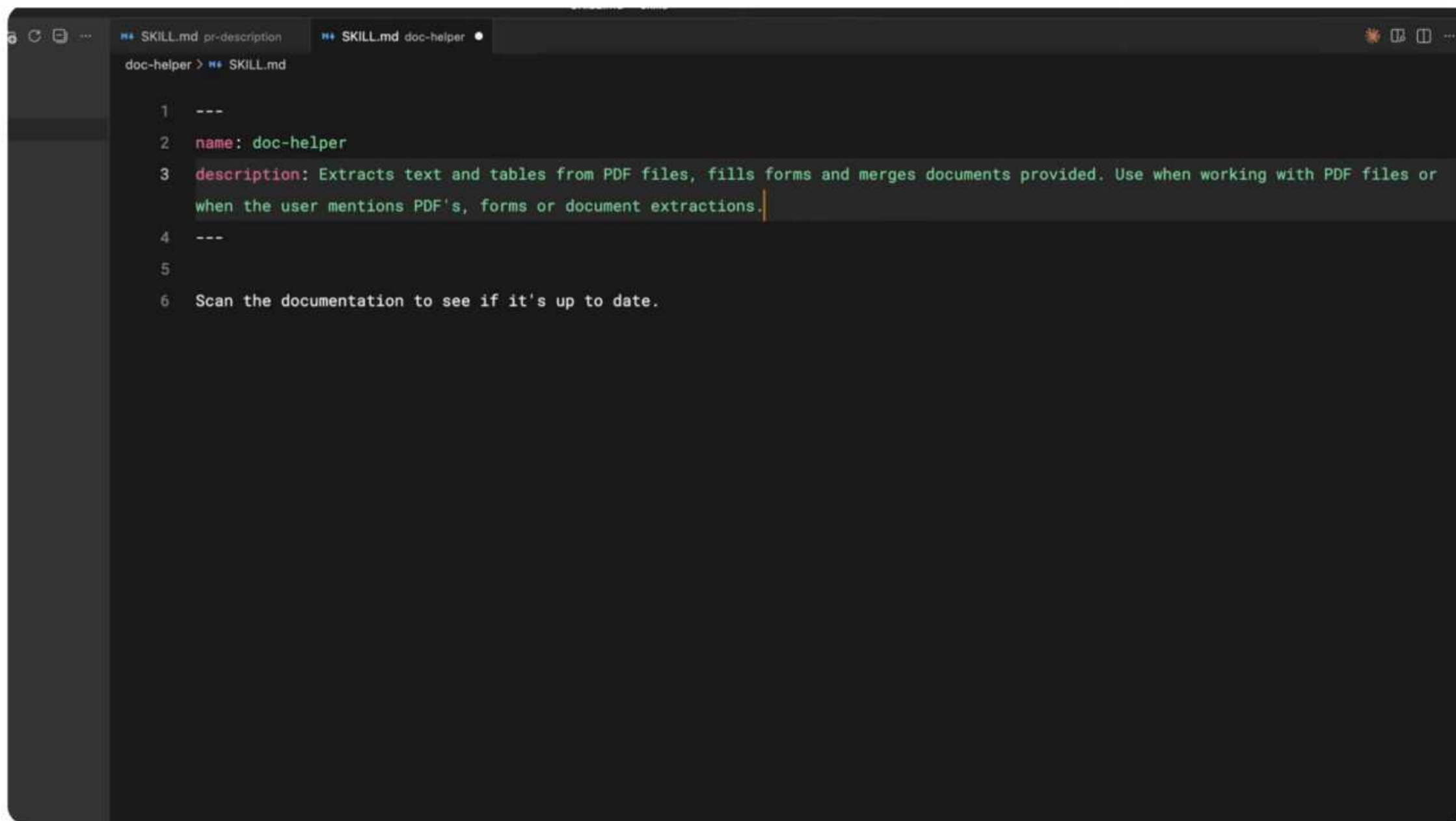


The image shows a VS Code editor window with two tabs: "SKILL.md pr-description" and "SKILL.md doc-helper X". The active tab is "SKILL.md doc-helper X". The editor content shows a command prompt prompt "doc-helper >" followed by the command "SKILL.md". Below the command, there is a list of lines from a file:

```
1 ---
2 name: doc-helper
3 description: Helps with docs
4 ---
5
6 Scan the documentation to see if it's up to date.
```

The text "Helps with docs" on line 3 is highlighted with a yellow selection box. The editor has a dark theme and a sidebar on the left.

Skill的描述

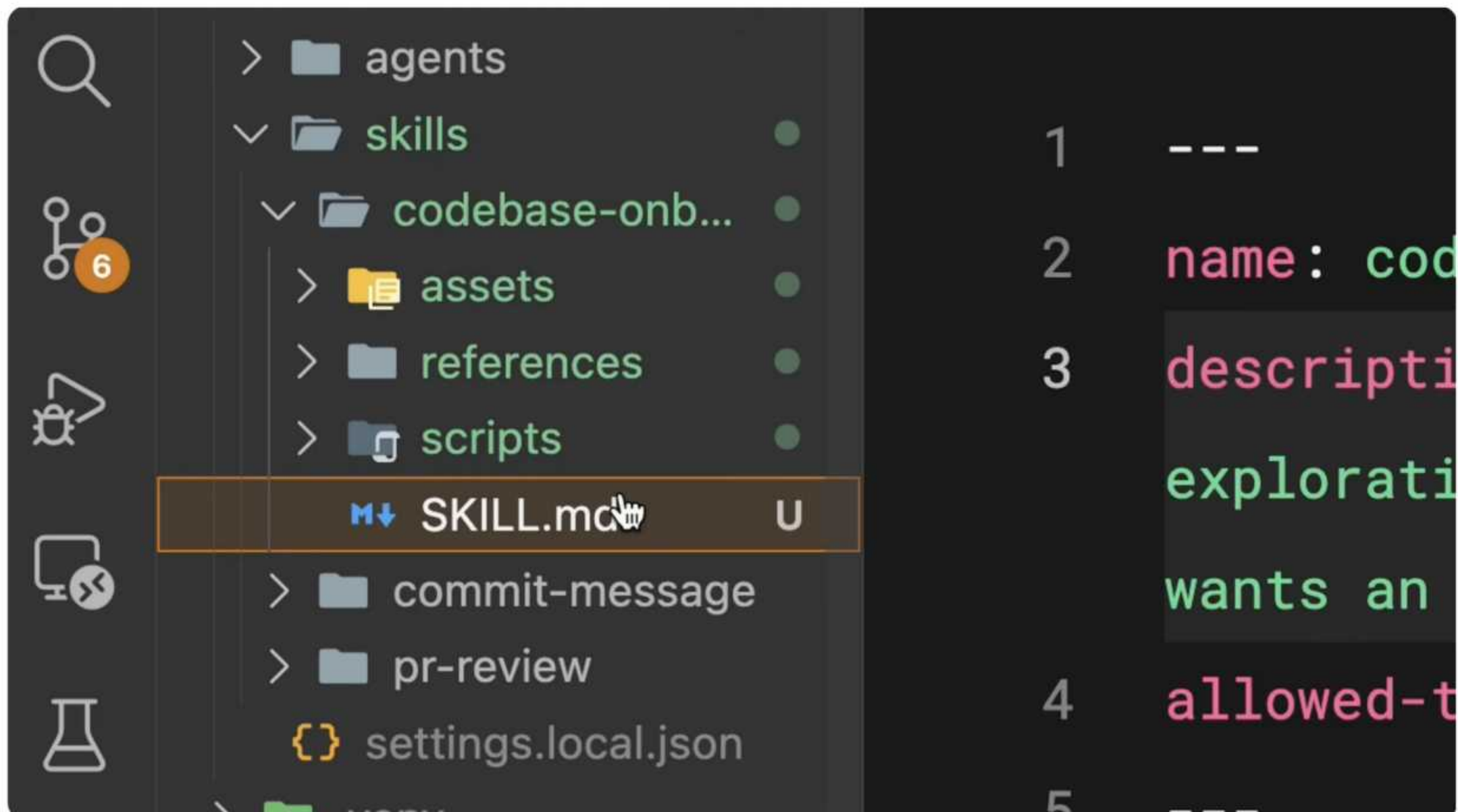


The image shows a code editor with two tabs: 'SKILL.md pr-description' and 'SKILL.md doc-helper'. The 'doc-helper' tab is active, showing a skill definition in a code block. The code is as follows:

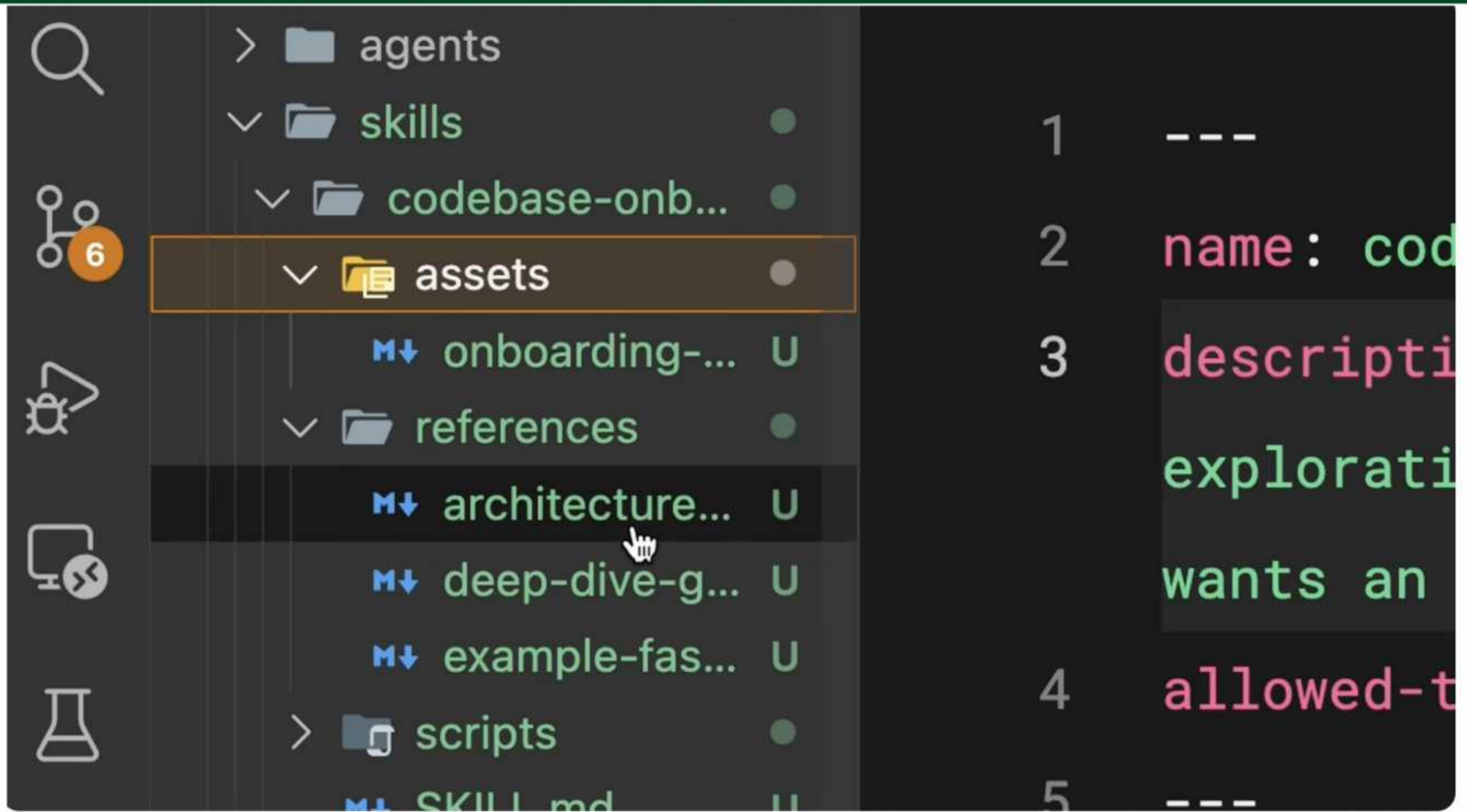
```
doc-helper > SKILL.md

1 ---
2 name: doc-helper
3 description: Extracts text and tables from PDF files, fills forms and merges documents provided. Use when working with PDF files or
  when the user mentions PDF's, forms or document extractions.
4 ---
5
6 Scan the documentation to see if it's up to date.
```

Skill的其他相关目录和文件



Skill的其他相关目录和文件



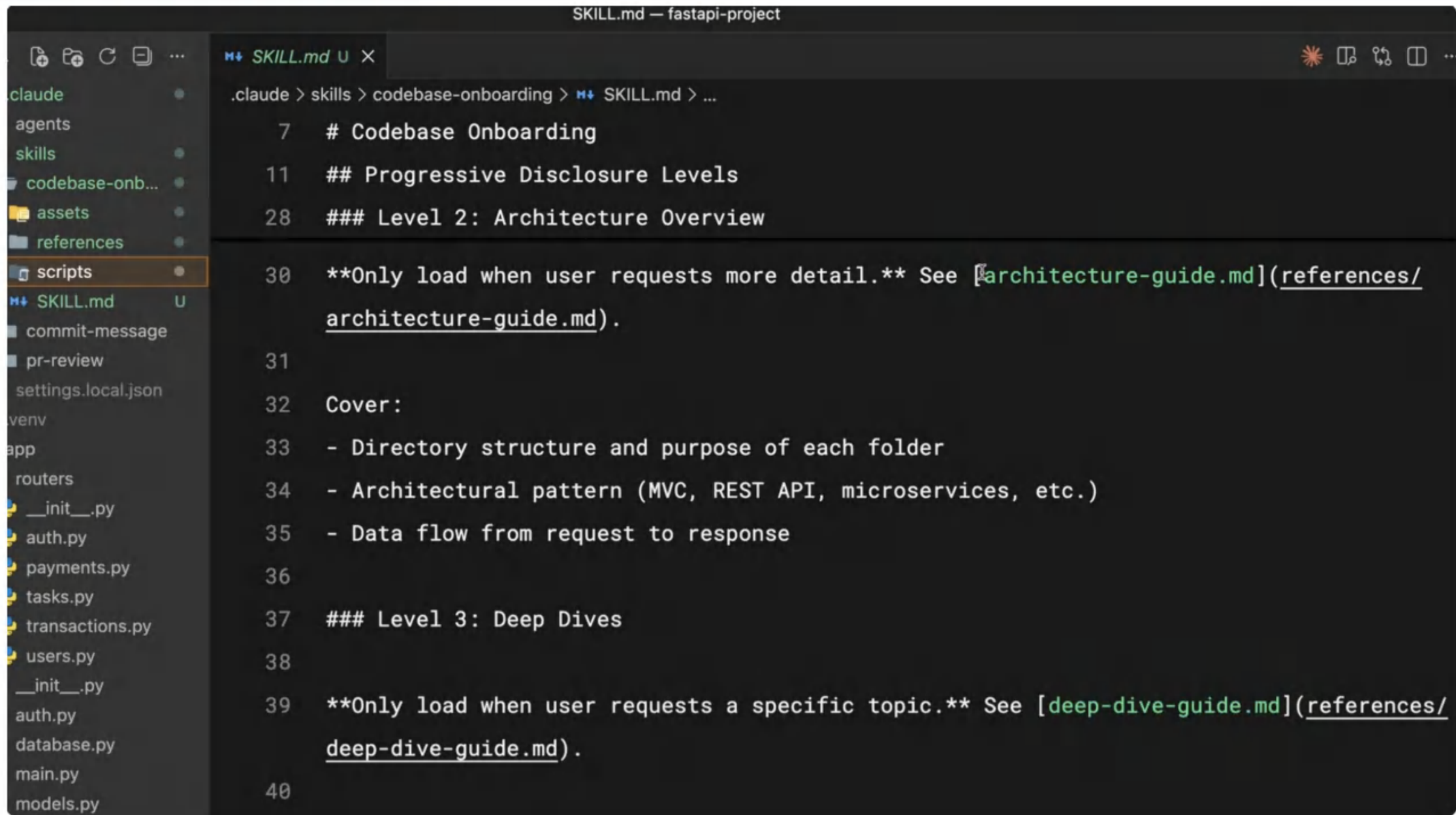
The image shows a file explorer on the left and a code editor on the right. The file explorer displays a directory structure with the following items:

- agents
- skills
- codebase-onb...
- assets** (highlighted with an orange box)
- onboarding-... U
- references
- architecture... U (with a mouse cursor pointing to it)
- deep-dive-g... U
- example-fas... U
- scripts
- SKILL.md

The code editor on the right shows a snippet of code with line numbers 1 through 5:

```
1 ---  
2 name: cod  
3 descripti  
   explorati  
   wants an  
4 allowed-t  
5 ---
```

Skill的reference



```
SKILL.md — fastapi-project

.claude > skills > codebase-onboarding > SKILL.md > ...

7   # Codebase Onboarding
11  ## Progressive Disclosure Levels
28  ### Level 2: Architecture Overview

30  **Only load when user requests more detail.** See [architecture-guide.md](references/
    architecture-guide.md).
31
32  Cover:
33  - Directory structure and purpose of each folder
34  - Architectural pattern (MVC, REST API, microservices, etc.)
35  - Data flow from request to response
36
37  ### Level 3: Deep Dives
38
39  **Only load when user requests a specific topic.** See [deep-dive-guide.md](references/
    deep-dive-guide.md).
40
```

Skill的脚本

diff main...

description t

e explaining

xt on why th

ints of spec

ated changes

ny files dele

fy this works

```
Codebase Onboarding x + v
2. New database table: Add model to models.py, add schemas to schemas.py
3. New field on existing model: Update model + schemas, tables auto-migrate

Useful Scripts

The skill includes helper scripts you can run:

# Check your dev environment
python .claude/skills/codebase-onboarding/scripts/check-setup.py

# List all API endpoints
python .claude/skills/codebase-onboarding/scripts/list-endpoints.py

---
Would you like me to:
- Run the setup check script to verify your environment?
- Show you a specific part of the codebase in detail?
- Explain how a particular feature (auth, tasks, payments) works?

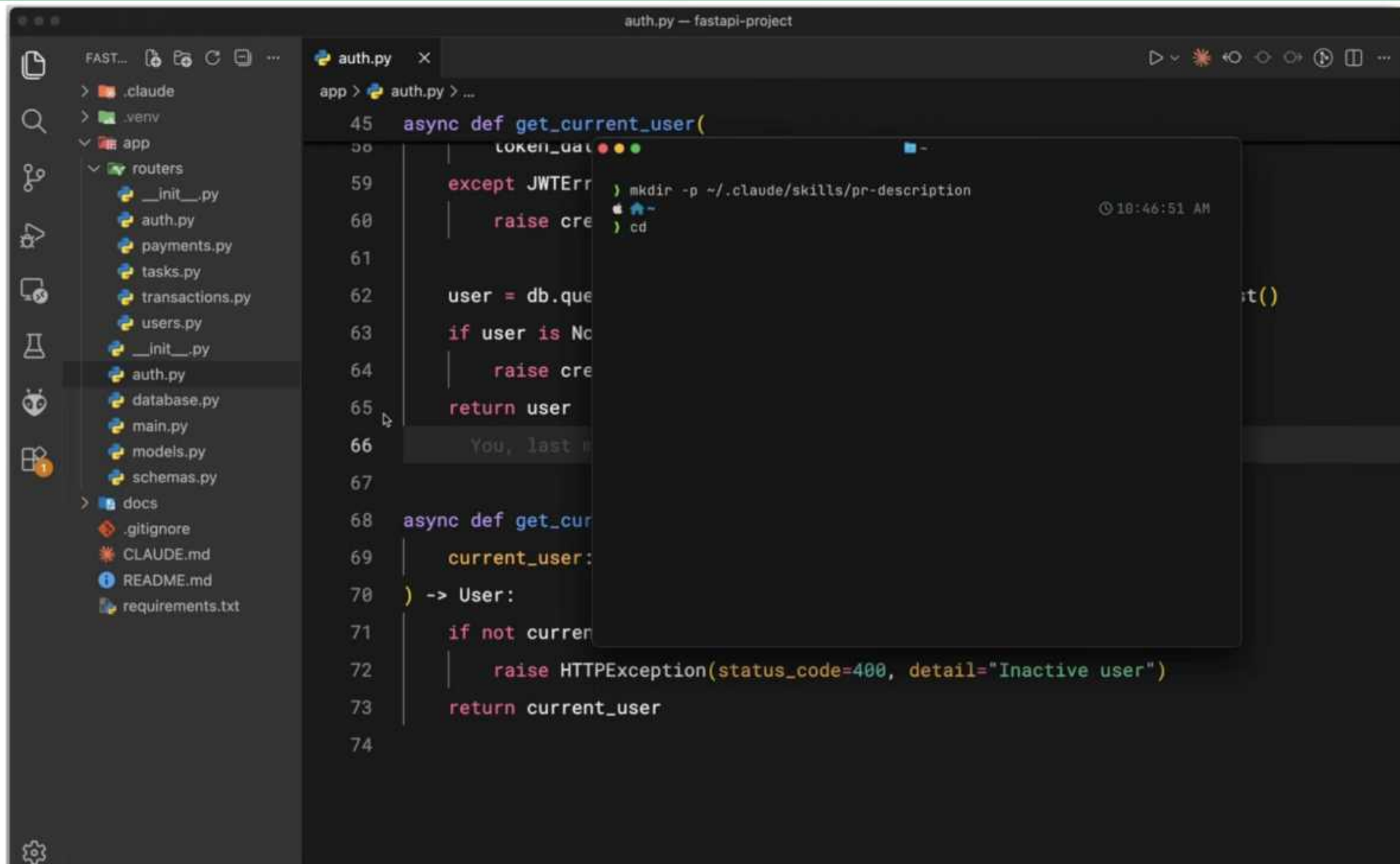
> Run the setup check script

* Cascading...
  L Tip: Run /install-github-app to tag @claude right from your Github issues and PRs

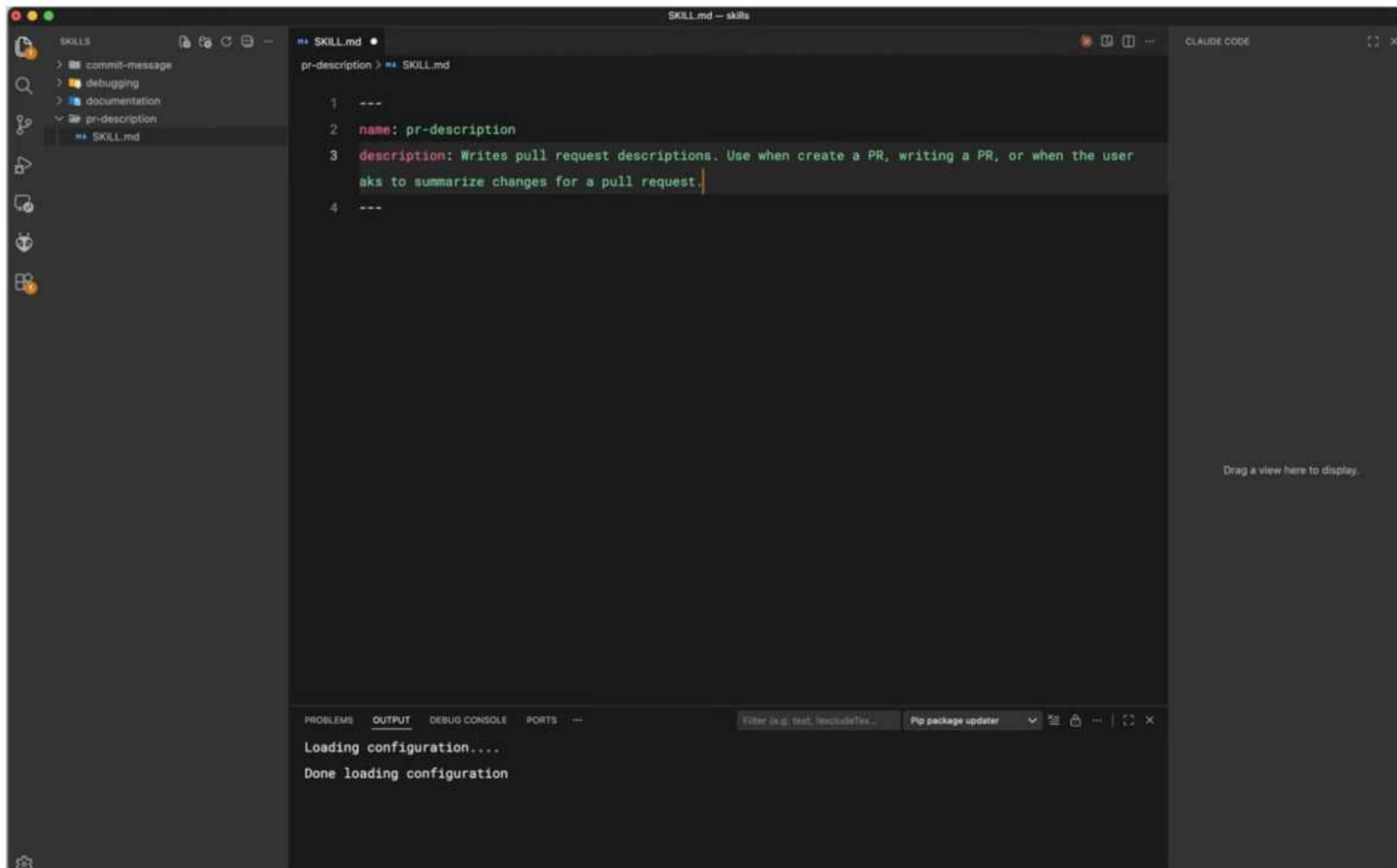
> █

esc to interrupt
```

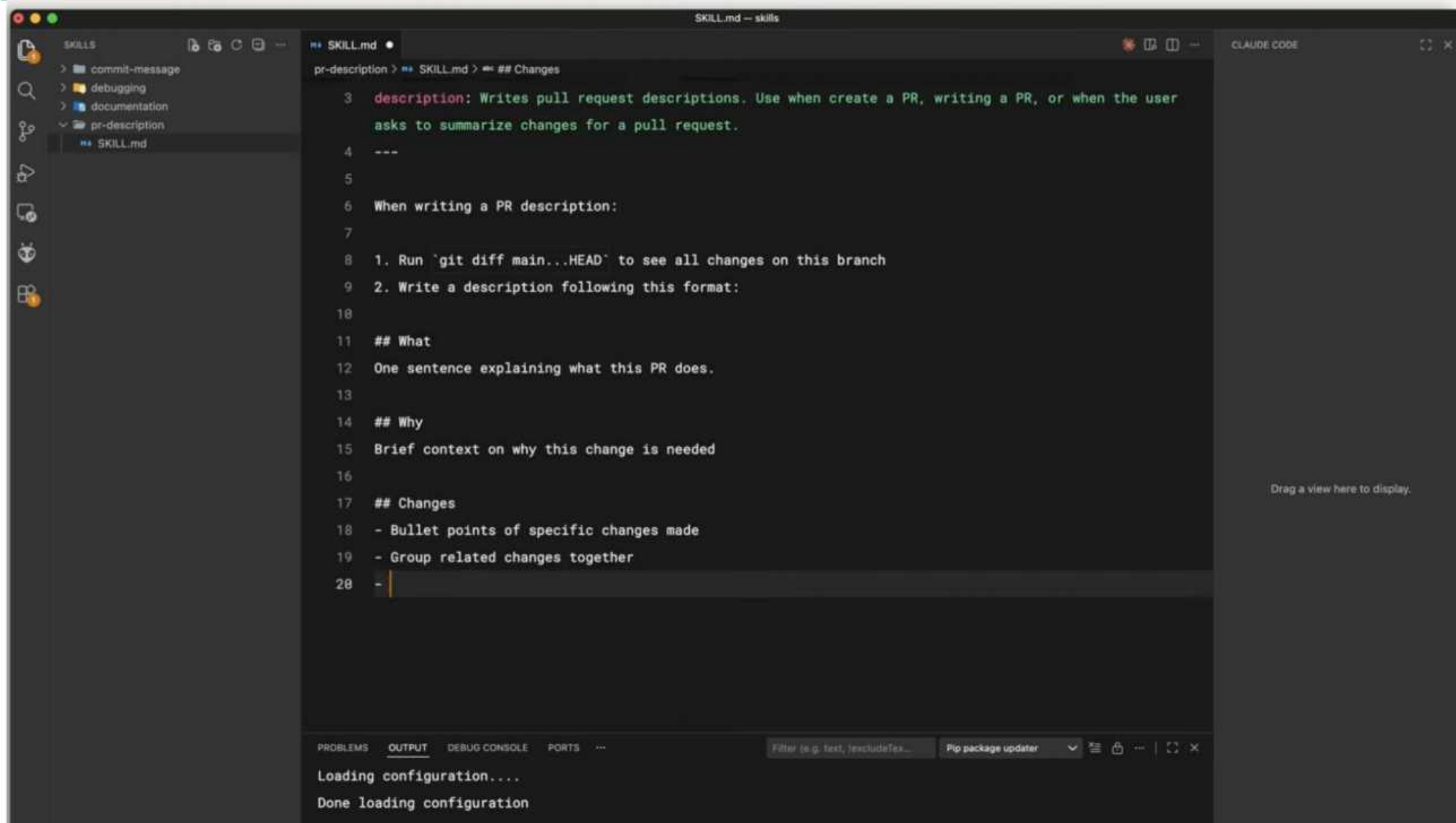
动手创建一个skill



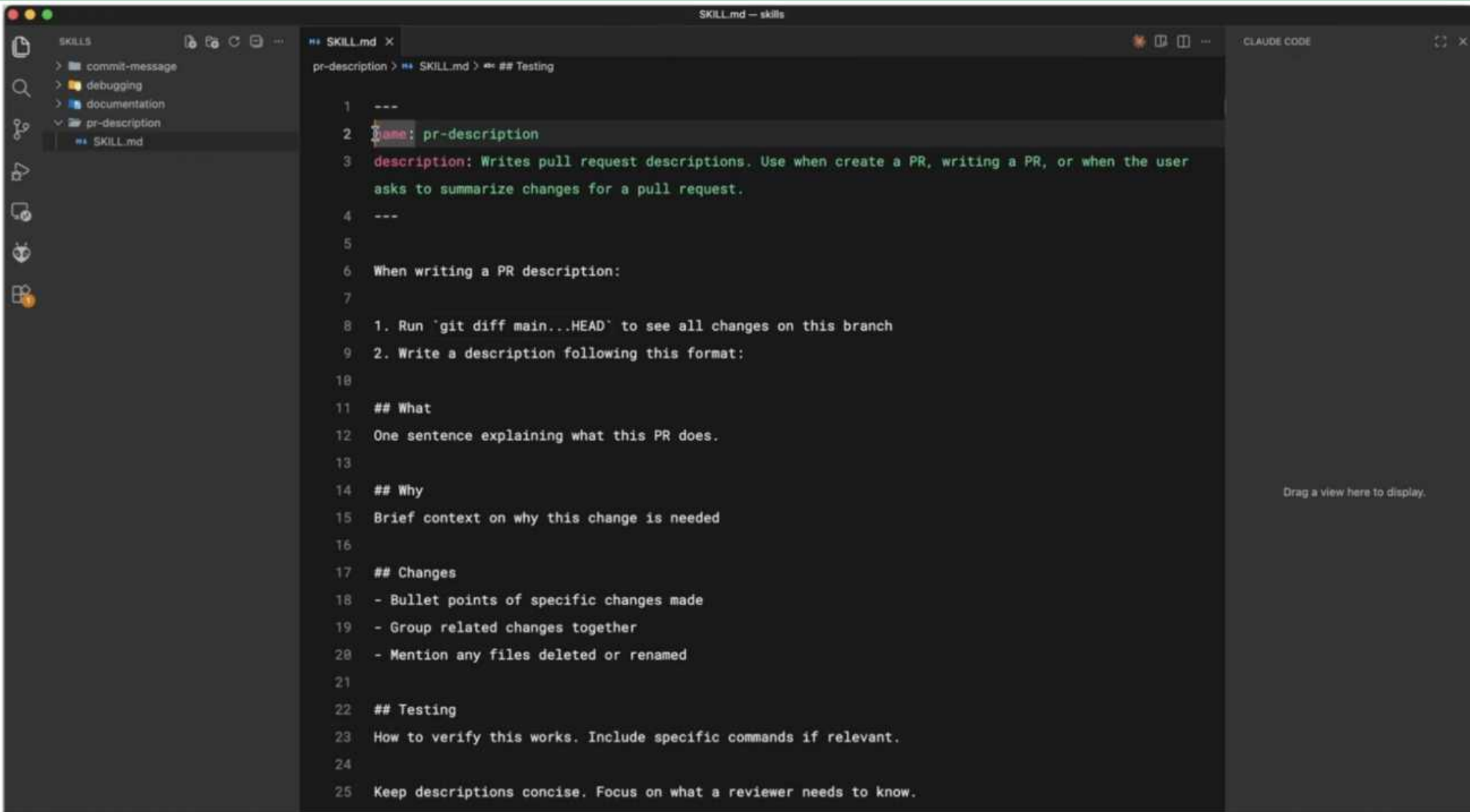
填写相应的内容



填写相应的内容



填写相应的内容



```
pr-description > SKILL.md > ## Testing

1 ---
2 name: pr-description
3 description: Writes pull request descriptions. Use when create a PR, writing a PR, or when the user
  asks to summarize changes for a pull request.
4 ---
5
6 When writing a PR description:
7
8 1. Run `git diff main...HEAD` to see all changes on this branch
9 2. Write a description following this format:
10
11 ## What
12 One sentence explaining what this PR does.
13
14 ## Why
15 Brief context on why this change is needed
16
17 ## Changes
18 - Bullet points of specific changes made
19 - Group related changes together
20 - Mention any files deleted or renamed
21
22 ## Testing
23 How to verify this works. Include specific commands if relevant.
24
25 Keep descriptions concise. Focus on what a reviewer needs to know.
```

查看skills

u, 2 days ago | 1 author

```
from datetime import
```

```
from typing import
```

```
from fastapi import
```

```
from fastapi.security
```

```
from jose import
```

```
from passlib.context
```

```
from sqlalchemy.orm
```

```
from app.database
```

```
from app.models import
```

```
from app.schemas
```

Available Skills

claude



Claude Code v2.1.20
Opus 4.5 · Claude Max
~/fastapi-project

What skills are available?

Hyperspacing... (esc to interrupt · thought for 1s)

? for shortcuts

○ IDE disconnected

相看skills

u, 2 days ago | 1 author

```
from datetime import
```

```
from typing import
```

```
from fastapi import
```

```
from fastapi.security
```

```
from jose import
```

```
from passlib.context
```

```
from sqlalchemy.c
```

```
from app.database
```

```
from app.models import
```

```
from app.schemas
```

Available Skills

bindings, or modify keybindings

Skill: documentation
Description: Write documentation - READMEs, API docs, code documentation

Skill: debugging
Description: Help debug issues, troubleshoot errors, fix bugs

Skill: pr-description
Description: Write pull request descriptions and summarize changes

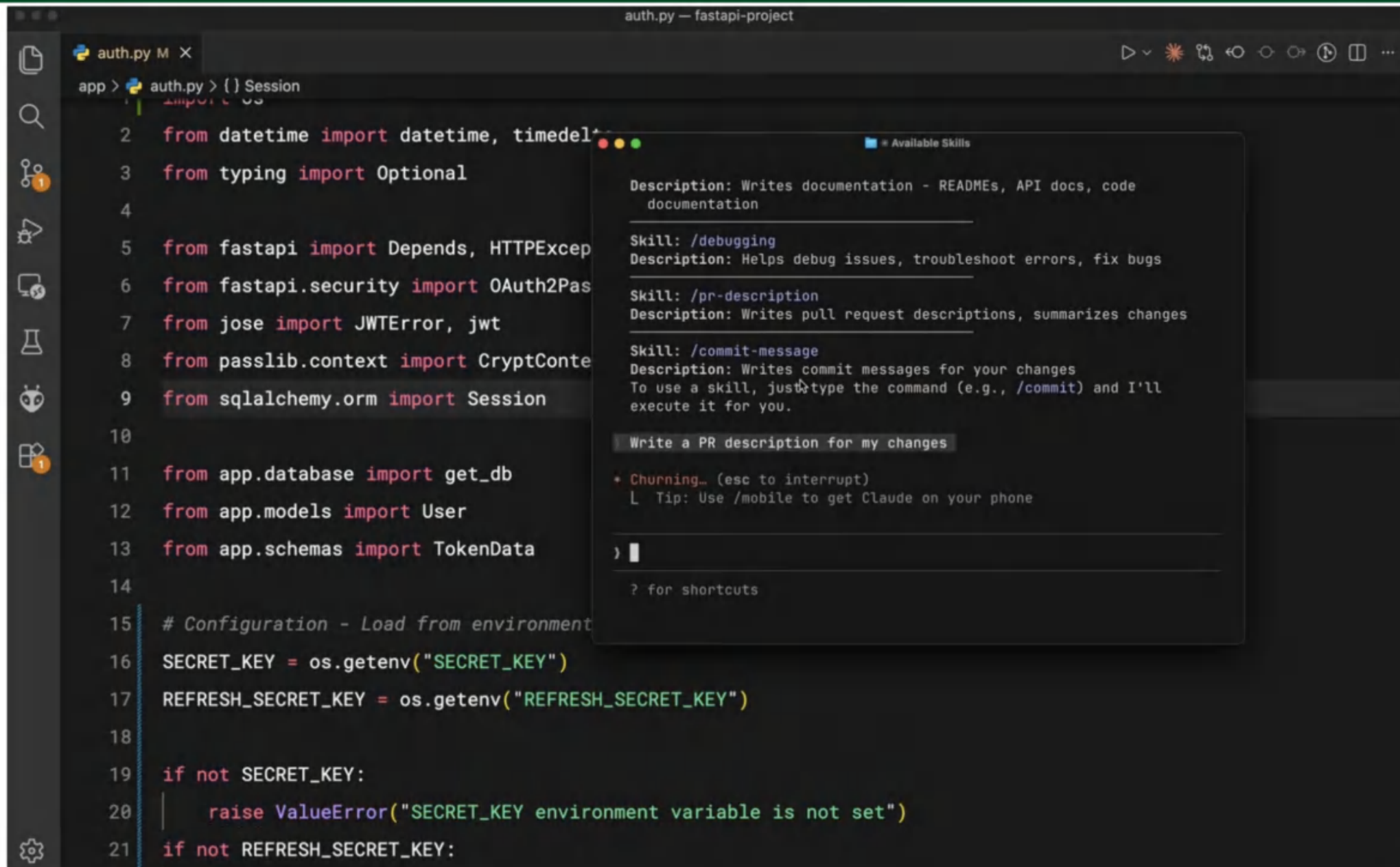
Skill: commit-message
Description: Write commit messages for staged changes
You can invoke these with a slash command like /commit or /debugging, or ask me to use them directly.

> |

? for shortcuts

○ IDE disconnected

调用skill



The image shows a VS Code editor window with a file named `auth.py` open. The file contains Python code for an authentication system using FastAPI, SQLAlchemy, and JWT. The code is as follows:

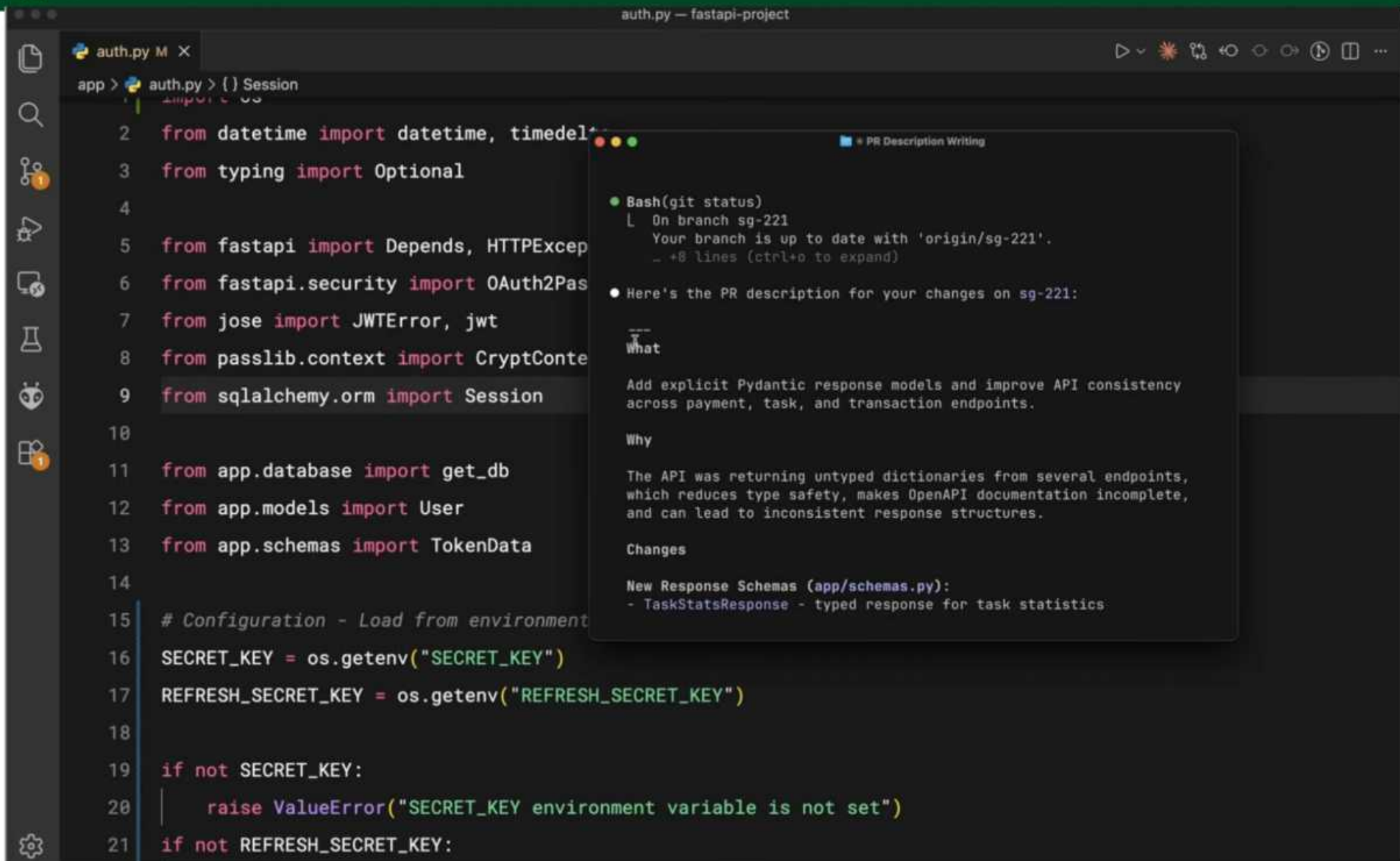
```
1 import os
2 from datetime import datetime, timedelta
3 from typing import Optional
4
5 from fastapi import Depends, HTTPException
6 from fastapi.security import OAuth2PasswordBearer
7 from jose import JWTError, jwt
8 from passlib.context import CryptContext
9 from sqlalchemy.orm import Session
10
11 from app.database import get_db
12 from app.models import User
13 from app.schemas import TokenData
14
15 # Configuration - Load from environment
16 SECRET_KEY = os.getenv("SECRET_KEY")
17 REFRESH_SECRET_KEY = os.getenv("REFRESH_SECRET_KEY")
18
19 if not SECRET_KEY:
20     raise ValueError("SECRET_KEY environment variable is not set")
21 if not REFRESH_SECRET_KEY:
```

On the right side of the editor, the 'Available Skills' panel is open, displaying a list of skills and their descriptions:

- Skill: /documentation**
Description: Writes documentation - READMEs, API docs, code documentation
- Skill: /debugging**
Description: Helps debug issues, troubleshoot errors, fix bugs
- Skill: /pr-description**
Description: Writes pull request descriptions, summarizes changes
- Skill: /commit-message**
Description: Writes commit messages for your changes
To use a skill, just type the command (e.g., /commit) and I'll execute it for you.

Below the skills list, there is a text input field containing the command `Write a PR description for my changes`. At the bottom of the panel, there is a status bar indicating `+ Churning... (esc to interrupt)` and a tip: `Tip: Use /mobile to get Claude on your phone`.

执行skill



The image shows a code editor window titled "auth.py — fastapi-project". The code is in Python and defines a FastAPI application with authentication. A PR description overlay is visible on the right side of the editor.

```
auth.py M X
app > auth.py > {} Session
1
2 from datetime import datetime, timedelta
3 from typing import Optional
4
5 from fastapi import Depends, HTTPException
6 from fastapi.security import OAuth2PasswordBearer
7 from jose import JWTError, jwt
8 from passlib.context import CryptContext
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19 if not SECRET_KEY:
20     raise ValueError("SECRET_KEY environment variable is not set")
21 if not REFRESH_SECRET_KEY:
```

PR Description Writing

- Bash(git status)
 - └ On branch sg-221
 - Your branch is up to date with 'origin/sg-221'.
 - +8 lines (ctrl+o to expand)
- Here's the PR description for your changes on sg-221:

What

Add explicit Pydantic response models and improve API consistency across payment, task, and transaction endpoints.

Why

The API was returning untyped dictionaries from several endpoints, which reduces type safety, makes OpenAPI documentation incomplete, and can lead to inconsistent response structures.

Changes

New Response Schemas (app/schemas.py):

- TaskStatsResponse - typed response for task statistics

什么是 harness?

> **Harness** 是包裹在 LLM 外面的执行系统，负责把用户任务转化为模型输入，把模型输出转化为工具调用、文件修改、代码执行、验证反馈和最终结果。

OpenAI 对 Codex 的解释很接近这个定义：Codex harness 提供核心的 **agent loop** 和执行逻辑，负责组织用户、模型和工具之间的交互。模型可以请求工具调用，harness 执行工具，再把工具结果放回上下文，循环直到任务结束。

([OpenAI][2])

什么是 harness?

Harness=(Prompting,Context,Tools,Memory,Execution,Validation,Recovery,Permissions)

模型看到什么上下文;
什么时候加载 skill;
可以调用哪些工具;
工具输出如何反馈给模型;
是否能运行代码、测试、lint、type checker;
是否能读写文件;
失败后是否自动重试;
最终结果如何验证。

为什么同一个skill 在不同 harness中效果不同??

因为 **skill** 通常只是“知识包”，但它需要 **harness** 来完成以下关键动

- **skill** 触发机制不同

- 工具接口不同

- 观察反馈格式不同

- 验证机制不同

- 失败恢复能力不同

GPT-5.5

□因此，严格地说，不应该说：

□> **GPT-5.5** 在某任务上成功率是多少。

□而应该说：

□> **GPT-5.5 + Skill X + Codex harness + 某工具环境
+ 某 budget** 的成功率是多少。

如何写好的 skill?

- **Name:** skill 名字要短、明确。
- **Trigger:** 什么时候使用, 什么时候不要使用。
- **Goal:** 这个 skill 解决什么问题。
- **Procedure:** 分步骤流程。
- **Tools:** 推荐使用哪些命令、脚本、库。
- **Constraints:** 不能做什么, 必须保留什么。
- **Validation:** 如何判断完成。
- **Failure Cases:** 常见错误与修复方式。

写 skill 的高级原则

□skill 的 **description** 比正文更重要

- ✓因为很多 harness 先加载 skill 的 metadata，再决定是否读取完整 skill。Anthropic 的 skill 机制就是先把 name 和 description 作为轻量信息加载，只有判断相关时才读完整 SKILL.md

□skill 要短主干、多文件展开

- ✓不要把所有知识放进一个巨大 SKILL.md

□skill 要包含“不要做什么”

□skill 要包含 **output contract**

- ✓返回 LaTeX；保留 equation 环境；不要使用 bullet；控制在 1000 字左右；输出修改后的完整段落，而不是解释。

□skill 要有自验证步骤

如何写好的 harness?

- ❑ **Planner**: 任务规划器
- ❑ **Context Manager**: 上下文管理器
- ❑ **Skill Router**: skill 路由器
- ❑ **Tool Executor**: 工具执行器
- ❑ **State Tracker**: 状态跟踪器
- ❑ **Verifier**: 验证器
- ❑ **Recovery Manager**: 错误恢复器

好 harness 的设计原则

❑ **harness** 必须把“想法”绑定到“状态”

✓ Before final answer:

- ❑ 1. List changed files.
- ❑ 2. Run required validators.
- ❑ 3. Confirm output artifact exists.
- ❑ 4. Compare result with user requirement.
- ❑ 5. Only then answer.

❑ **harness** 要有强 **output contract**

❑ **harness** 要让工具输出可学习

❑ **harness** 要内置验证循环

❑ **harness** 要管理权限和风险

❑ **harness** 要支持 **trace**

多数情况下，不需要从零写 harness

- 目标是解决实际问题，例如写代码、改论文、生成 **PPT**、处理 **repo**、跑测试、做数据分析，那么直接用现成 **harness** 就可以

什么时候需要自己写一点 harness

- 研究 **agent skill** 的效果，那么只用现成 **harness** 还不够。因为 **skill** 的效果会被 **harness** 强烈影响
- thin harness wrapper**
- 最小 harness**

什么时候真正写harness

□如果研究重点是 **harness**，才需要真正写 **harness**

最小 harness

```
mini_skill_harness/  
  skills/  
    code-review/  
      SKILL.md  
    latex-refine/  
      SKILL.md  
    bug-fix/  
      SKILL.md  
  
  tasks/  
    tasks.jsonl
```

```
harness.py  
runners/  
  codex_runner.py  
  claude_runner.py  
  
verifiers/  
  run_tests.py  
  check_diff.py  
  check_artifact.py  
  
logs/  
  runs.jsonl
```

最小harness

```
import asyncio
from claude_agent_sdk import query, ClaudeAgentOptions

async def run_claude(prompt: str, workdir: str):
    options = ClaudeAgentOptions(
        allowed_tools=["Read", "Edit", "Bash"],
        cwd=workdir,
    )

    messages = []
    async for message in query(prompt=prompt, options=options):
        messages.append(message)

    return messages

asyncio.run(run_claude("Fix the failing test.", "./repo"))
```

Many agent skill papers are not fundamentally innovative.

- ❑ skill library = prompt/code memory**
- ❑ skill acquisition = LLM reflection + edit + retry**
- ❑ skill selection = retrieval or ranking**
- ❑ skill evolution = prompt optimization over trajectories**

Native Agent Skill Model

$$\mathcal{S}_i = (z_i, \pi_i, C_i, E_i, R_i, V_i) \quad \text{a skill}$$

$$p_i = C_i(s_t, g) \quad \text{is the precondition}$$

$$i^* = \arg \max_i C_i(s_t, g) [V_i(s_t, g) - \lambda R_i(s_t, g)]$$

the success/value estimator the resource cost

More innovative contribution: skill algebra

The real novelty could be to define how skills compose:

$$\mathcal{S}_{ij} = \mathcal{S}_j \circ \mathcal{S}_i$$

with the condition:

$$E_i(s) \in \text{Dom}(C_j)$$

一些前沿方向

- 从静态获取走向动态的技能进化
- 资源感知与多目标选择
- 多模态与物理世界的技能落地

SkillOpt

- s : 技能文档，一段纯文本指令，例如一段 Markdown。
- M : 冻结的目标执行模型，例如 GPT-5.5。
- O : 独立的优化器模型，用于反思和修改技能。
- x : 具体的任务，例如“计算 A 列的总和并填入 B1”。
- h : 执行框架，负责将模型、任务和技能结合执行。
- $\tau(s)$: 执行轨迹，包含模型的思考过程、代码生成、报错信息等。
- $r(s)$: 任务得分， $r(s) \in [0, 1]$ ，其中 1 代表完全成功，0 代表失败。

第 1 步：前向传播与收集 Rollout 证据

$$(\tau(s_{cur}), r(s_{cur})) = h(M, x, s_{cur})$$

- 当前技能 s_{cur} ：使用 Python 的 pandas 库来处理电子表格。
- 执行：模型 M 在 40 个电子表格任务上执行该技能。
- 结果：25 个任务成功，得分 $r = 1$ ；15 个任务失败，得分 $r = 0$ 。在失败轨迹 τ 中，报错信息显示：测试系统期望读取单元格的 **静态数值**，但模型却写入了 Excel 公式，例如 **INDEX/MATCH**，导致判分系统无法读取。

第 2 步：小批量反思

$$E_{fail} = O(\mathcal{B}_{fail}, s_{cur})$$

$$E_{succ} = O(\mathcal{B}_{succ}, s_{cur})$$

- 优化器 O 拿到 15 个因为“写入公式而不是静态数值”而失败的轨迹。
- 优化器分析后发现这是一个系统性错误，于是生成一个 JSON 格式的修改建议。
- 操作： `append` ，表示追加文本。
- 内容： 检查工作表结构和公式后，请将计算出的静态数值（evaluated static values）写入目标范围，而不要依赖 Excel 的公式重算功能。

第 3 步：应用有界文本更新，也就是文本学习率

$$E_{top} = \text{TopK}(E_{merged}, L_t)$$

$$\tilde{s} = \text{Apply}(s_{cur}, E_{top})$$

- 当前的“学习率”预算为 $L_t = 4$ 。
- 由于“要求写入静态数值”的建议能够修复大量系统性失败，优化器将其排在第 1 位。
- 系统将这条建议应用到文档中，生成 候选技能 $\tilde{s}$ ：

“使用 Python 的 pandas 库来处理电子表格。 新增：检查工作表结构和公式后， 请将计算出的静态数值写入目标范围， 而不要依赖 Excel 的公式重算功能。”

第 4 步：验证门控

$$score_{cand} = \frac{1}{|D_{sel}|} \sum_{x \in D_{sel}} r_x(\tilde{s})$$

If $score_{cand} > score_{cur}$, then $s_{cur} \leftarrow \tilde{s}$

- 我们拿着候选技能 $\tilde{s}$ 在独立的 D_{sel} 数据集上测试。
- **测试结果**：旧技能的正确率是 **40%**；新技能由于明确指出了“静态数值”的要求，正确率提升到 **55%**。
- **决策**：因为 $0.55 > 0.40$ ，所以验证通过。这条针对 Spreadsheet 的修改被正式永久写入技能文档。

Residual Learning

□ Use student learning example

问题和讨论

